

AC7925A Datasheet

Zhuhai Jieli Technology Co.,LTD

Version 1.1

Date 2024.09.30

Copyright © Zhuhai Jieli Technology Co.,LTD. All rights reserved.

Revision History

Date	Revision	Author	Description
2024.05.30	V1.0	zh-jieli	Initial Release
2024.09.30	V1.1	zh-jieli	Add Power Domain Information

Table of Contents

Revision History	1
Table of Contents	2
AC7925A Features	3
1 Block Diagram	5
2 Pin Definition	6
2.1 Pin Assignment	6
2.2 Pin Description	7
3 Electrical Characteristics	10
3.1 Absolute Maximum Ratings	10
3.2 ESD Ratings	10
3.3 PMU Characteristics	10
3.4 IO Characteristics	11
3.5 Audio DAC Characteristics	13
3.6 Audio ADC Characteristics	14
3.7 BT Characteristics	15
3.8 WiFi Characteristics	15
4 Package Information	17
4.1 QFN68_7*7mm	17
5 IC Marking Information	18
6 Solder-Reflow Condition	19

AC7925A Features

SYSTEM

- Dual Core 32bit DSP 320MHz
- With IEEE754 Single precision FPU
- Support jieli TEE
- Support FFT / MATRIX / MATH
- 2 x I-cache and D-cache
- On-chip SRAM 352kbyte
- Support SDTAP / EMU / ETM
- Support MMU
- Support MPU
- Built-in SDRAM/DDR (Maximum 64Mbyte)
- SPI FLASH Controller (Maximum 64Mbyte)
- 24MHz crystal oscillator
- 32KHz RTC crystal oscillator
- Internal RC oscillator, PLL

Video Input

- Internal Image Signal Processor
- Support DVP, BT656, SPI interface
- Support RAW, YUV422 formats
- Support video resize an time mark
- 2 x JPEG codec

Video output

- Support display color enhancement
- Support DPI, DBI, BT656 interface
- Support 4 lane MIPI-DSI interface
- Support RGB, YUV formats

Graphics

- Internal 2D DMA
- Internal 2.5D GPU
- Support vector graphics rendering
- Support image resize, rotation, projection
- Support multiple blending mode
- Support ARGB, RGB, YUV, Lx, Ax formats

DSP Audio Processing

- SBC/AAC/LDAC/LHDC/LC3/CVSD/mSBC codec
- mSBC voice codec supported for BT phone
- PLC for voice processing

- Single/Multi MIC ENC
- Multi-band DRC
- Multi-band EQ
- Support spatial sound

Audio

- 2 x 16bit DAC
 - ❖ SNR 103dB
 - ❖ Noise 6.4uVrms
 - ❖ Supports differential mode
 - ❖ Sampling rate 8~96kHz
- 2 x 16bit ADC
 - ❖ SNR 95dB
 - ❖ Sampling rate 8~48kHz
- I2S/PDM AUDIO Master/Slave interface

Bluetooth

- Dual-mode BT5.4 with LE Audio (QDID 223418)
- Support AoA/AoD
- Support LE audio BIS/CIS
- Support long range BLE
- Maximum transmitting power 19 dBm
- Receiver sensitivity
 - ❖ -95.5 dBm @BR
 - ❖ -96 dBm @EDR $\pi/4$ DQPSK
 - ❖ -88 dBm @EDR 8DPSK

IEEE 802.11b/g/n

- 1T1R in 2.4 GHz band
- 20 MHz and 40 MHz bandwidth
- Data rate up to 150 Mbps
- Security:WFA/WPA3 personal,WPS2.0,WAPI
- QoS: WFA WMM, WMM PS
- Support STBC, A-MPDU, A-MSDU, BLK-ACK
- Support Station, SoftAP,Station+SoftAP, Promiscuous mode
- Maximum transmitting power
 - ❖ 19 dBm @1Mbps, DSSS
 - ❖ 17 dBm @HT20, MCS0
 - ❖ 13 dBm @HT20, MCS7

- Receiver sensitivity
 - ❖ -97 dBm @1Mbps, DSSS
 - ❖ -93 dBm @HT20, MCS0
 - ❖ -74 dBm @HT20, MCS7

Peripherals

- 1 x High speed USB
- 2 x SD host controller
- 6 x Multi-function 32bit timer
- 5 x UART interface
- 3 x I²C Master/Slave interface
- 3 x SPI Master/Slave interface
- 1 x QDEC
- 1 x CAN Controller
- 1 x Ethernet Mac
- 8 x MCPWM
- 3 x Light strip Controller
- 1 x 10bit ADC(4 Channel)
- 29 x GPIO Support function remapping
- Built-in RTC with alarm, wakeup

PMU

- 1 x Buck DC-DC converter
- 3 x IO power domain
- WIFI PA support external power supply
- RTCVDD33 support external power supply
- Support temperature sensor
- VBAT range 2.7V to 5.5V
- IOVDD range 2.7V to 3.6V

Packages

- QFN68(7mm*7mm)

Temperature

- Operating temperature
 - TC = -20℃ to +85℃(standard range)
 - TC = -40℃ to +105℃(extended range)
- Storage temperature -65℃ to +150℃

Applications

- IPC
- Driving recorder
- WIFI Appliances

1 Block Diagram

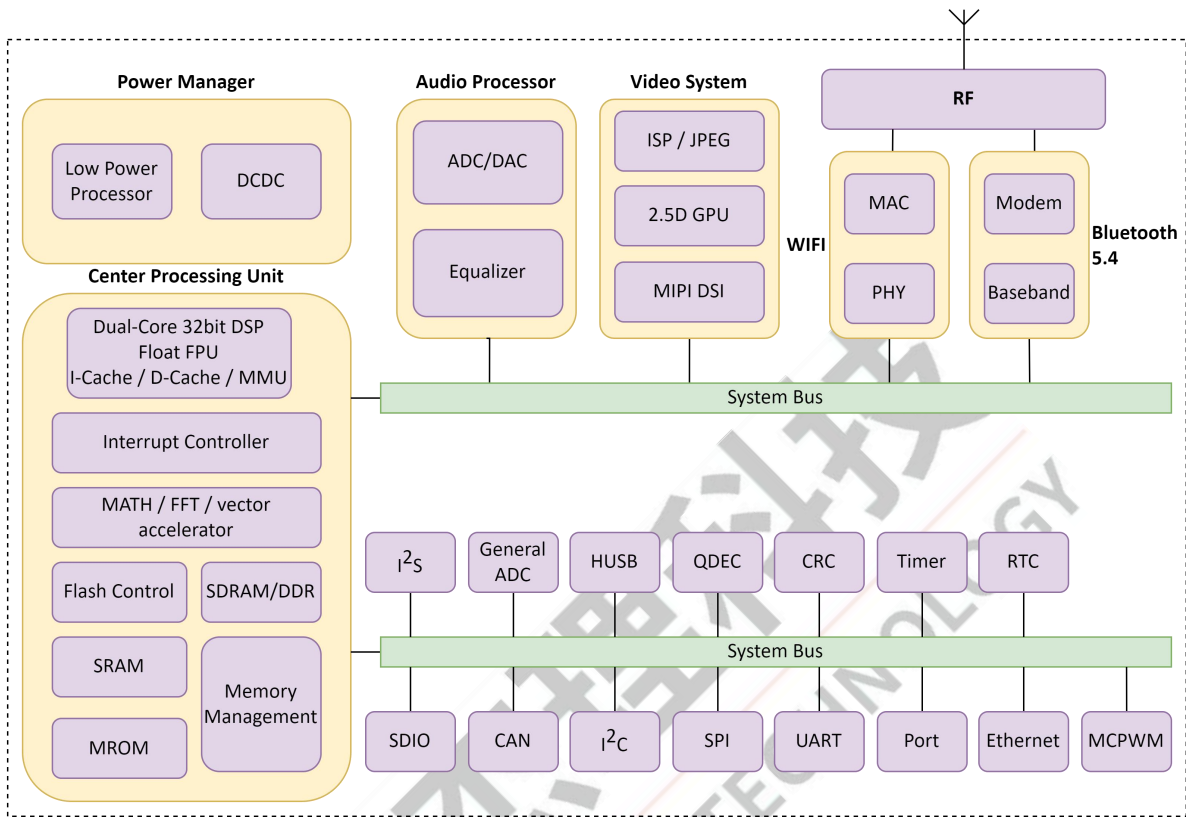


Figure 1-1 AC7925A Block Diagram

2 Pin Definition

2.1 Pin Assignment

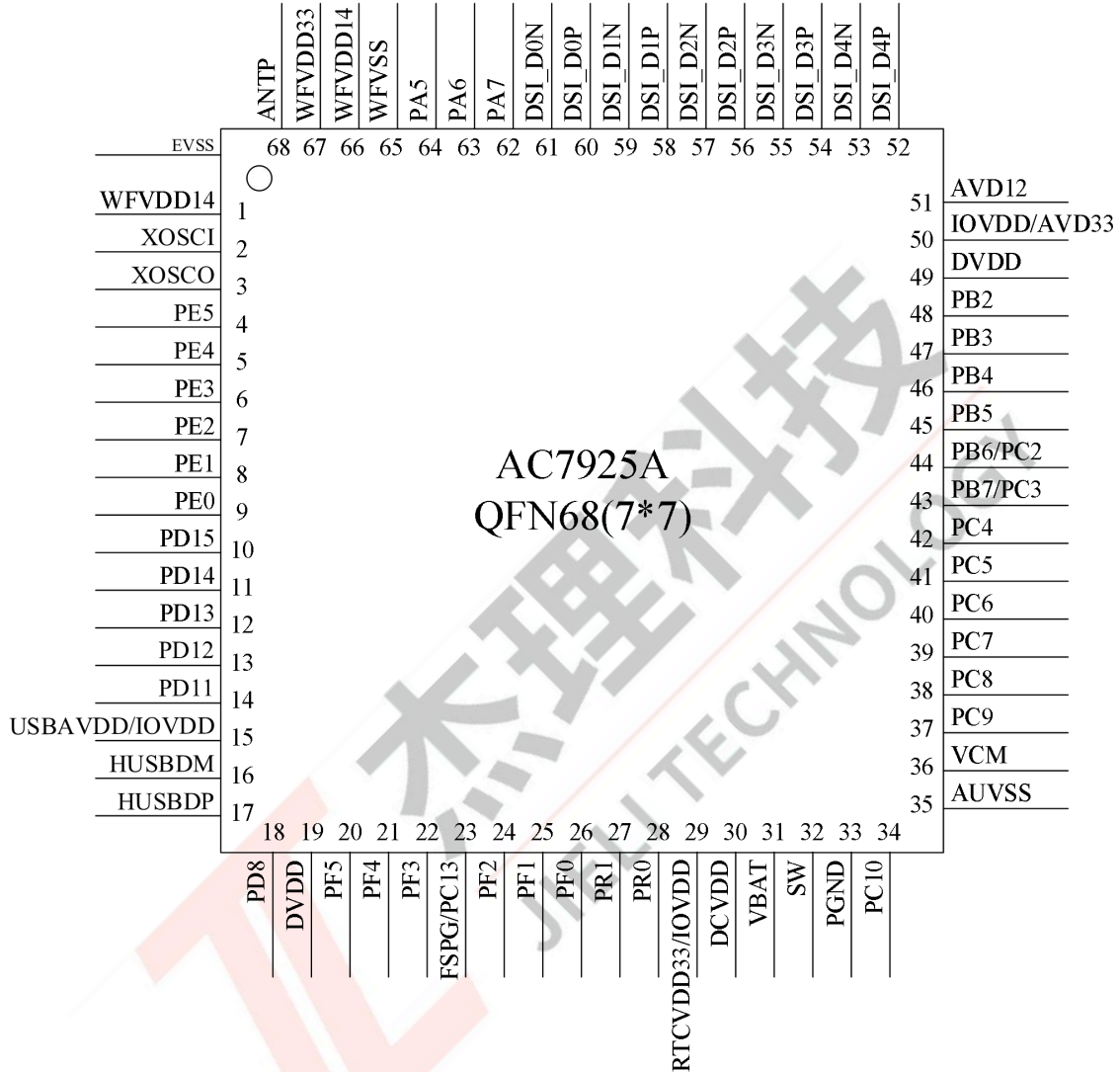


Figure 2-1 AC7925A Pin Assignment

2.2 Pin Description

Table 2-2-1 AC7925A Pin Description

Pin No.	Name	Type	IO Initial State	Description
1	WVDD14	P	--	Wireless 1.4V Power
2	XOSCI	I	--	Crystal Oscillator Input
3	XOSCO	O	--	Crystal Oscillator Output
4	PE5	I/O	Z	ADC8(ADC Input Channel 8) IO Wakeup Channel 8
5	PE4	I/O	Z	ADC7(ADC Input Channel 7) IO Wakeup Channel 7
6	PE3	I/O	Z	SensorX_CLK
7	PE2	I/O	Z	SensorX_D7 Ethnet RMII_RX1(A)
8	PE1	I/O	Z	SensorX_D6 Ethnet RMII_RX0(A)
9	PE0	I/O	Z	SensorX_D5 Ethnet RMII_CRSDV(A) SD1_CLK(B)
10	PD15	I/O	Z	SensorX_D4 Ethnet RMII_REFCLK(A) SD1_CMD(B)
11	PD14	I/O	Z	SensorX_D3 Ethnet RMII_RXERR(A) SD1_DATA0(B)
12	PD13	I/O	Z	SensorX_D2 Ethnet RMII_TXEN(A) SD1_DATA1(B)
13	PD12	I/O	Z	SensorX_D1 Ethnet RMII_TX1(A) SD1_DATA2(B)
14	PD11	I/O	Z	SensorX_D0 Ethnet RMII_TX0(A) SD1_DATA3(B)
15	USBAVDD	P	--	High Speed USB Power
	IOVDD	P	--	IO Power for PA5~PA7, PB2~PB7, PC2~PC10, PC13, PD8, PD11~PD15, PE0~PE5, PF0~PF5
16	HUSBDM	I/O	15kΩ Pull-down	High Speed USB Negative Data
17	HUSB DP	I/O	15kΩ Pull-down	High Speed USB Positive Data

Pin No.	Name	Type	IO Initial State	Description
18	PD8	I/O	10kΩ Pull-up	MCLR(Device Reset) ADC4(ADC Input Channel 4) SD Power IO Wakeup Channel 4
19	DVDD	P	--	Digital Logic Power
20	PF5	I/O	Z	SFCTZ_DO SPITZ_DO
21	PF4	I/O	Z	SFCTZ_CLK SPITZ_CLK
22	PF3	I/O	Z	SFCTZ_DATA3 SPITZ_DATA3
23	FSPG	I/O	Z	Flash Power Output
	PC13	I/O	Z	ADC13(ADC Input Channel 13) IO Wakeup Channel 13
24	PF2	I/O	Z	SFCTZ_DATA2 SPITZ_DATA2
25	PF1	I/O	Z	SFCTZ_DI SPITZ_DI
26	PF0	I/O	Z	SFCTZ_CS SPITZ_CS
27	PR1	I/O	Z	32k Crystal Oscillator Output
28	PR0	I/O	Z	32k Crystal Oscillator Input
29	RTCVDD33	P	--	RTC Power for PR0~PR1
	IOVDD	P	--	IO Power
30	DCVDD	P	--	DCDC Power
31	VBAT	P	--	Battery Input
32	SW	P	--	Buck DCDC Switch Port
33	PGND	G	--	Ground of Buck DC-DC converter
34	PC10	I/O	10kΩ Pull-down	LVD(External Low Voltage Detection Input) MICBIASB (MIC Bias Output)
35	AUVSS	G	--	Audio Ground
36	VCM	P	--	Audio Reference Power
37	PC9	I/O	Z	MICBIASA (MIC Bias Output) Right Channel DAC Output
38	PC8	I/O	Z	Left Channel DAC Output
39	PC7	I/O	Z	AIN_AP0(Audio ADC Positive Input)
40	PC6	I/O	Z	AIN_AN0(Audio ADC Negative Input)
41	PC5	I/O	Z	AIN_BN1(Audio ADC Negative Input)
42	PC4	I/O	Z	AIN_BP1(Audio ADC Positive Input)
43	PC3	I/O	Z	AIN_AP1(Audio ADC Positive Input)

Pin No.	Name	Type	IO Initial State	Description
	PB7	I/O	Z	SD0_CLK(D)
44	PC2	I/O	Z	AIN_AN1(Audio ADC Negative Input)
	PB6	I/O	Z	SD0_CMD(D)
45	PB5	I/O	Z	SD0_DATA0(D)
46	PB4	I/O	Z	SD0_DATA1(D)
47	PB3	I/O	Z	SD0_DATA2(D)
48	PB2	I/O	Z	SD0_DATA3(D)
49	DVDD	P	--	Digital Logic Power
50	IOVDD	P	--	IO Power
	AVD33	P	--	Analog 3.3V Power
51	AVD12	P	--	Analog 1.2V Power
52	DSI_D4P	I/O	--	MIPI DSI D4P
53	DSI_D4N	I/O	--	MIPI DSI D4N
54	DSI_D3P	I/O	--	MIPI DSI D3P
55	DSI_D3N	I/O	--	MIPI DSI D3N
56	DSI_D2P	I/O	--	MIPI DSI D2P
57	DSI_D2N	I/O	--	MIPI DSI D2N
58	DSI_D1P	I/O	--	MIPI DSI D1P
59	DSI_D1N	I/O	--	MIPI DSI D1N
60	DSI_D0P	I/O	--	MIPI DSI D0P
61	DSI_D0N	I/O	--	MIPI DSI D0N
62	PA7	I/O	Z	--
63	PA6	I/O	Z	--
64	PA5	I/O	Z	--
65	WVSS	G	--	Ground of Wireless
66	WVDD14	P	--	Wireless 1.4V Power
67	WVDD33	P	--	Wireless 3.3V Power
68	ANTP	RF	--	Antenna Positive Port

Note

- 1.IO initial state abbreviations Z--High resistance, H--High level, L--Low level, X--May be changed during power on.
- 2.Timer, CAN, MCPWM, QDEC, UART, LEDC, I²C, I²S and SPI functions can be remapped to any I/O (except PF/PR/DSI).

Table 2-2-2 Pin Types Description

Pin Type	Description	Pin Type	Description
P	Power	I/O	Input or Output
G	Ground	I	Input
RF	RF antenna	O	Output

3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1 Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Unit
T _{opt}	Operating temperature	-20	+85	°C
T _{stg}	Storage temperature	-65	+150	°C
VBAT	Supply Voltage	-0.3	5.5	V
IOVDD		-0.3	3.6	V
RTCVD33		-0.3	3.6	V
DCVDD		-0.3	1.54	V
WVDD33		-0.3	3.6	V
WVDD14		-0.3	1.54	V
USBAVDD		-0.3	3.6	V
AVD33		-0.3	3.6	V
AVD12		-0.3	1.54	V
GPIO	Input voltage of GPIO	-0.3	3.6	V

Note

1.Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device.

3.2 ESD Ratings

Table 3-2 ESD Ratings

Parameter	Typ	Test pin	Reference standard
Human Body Mode	±4kV	All pins	JEDEC EIA/JESD22-A114
Machine Mode	±300V	All pins	JEDEC EIA/JESD22-A115
Charge Device Model	±1kV	All pins	ANSI/ESDA/JEDEC JS-002-2022

3.3 PMU Characteristics

Table 3-3 PMU Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
VBAT	Power supply	--	2.7	3.7	5.5	V
Operating mode						
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
IOVDD	Voltage output	--	--	3.3	--	V
	Loading current	IOVDD=3.3V@VBAT = 3.9V	--	--	200	mA
DCVDD	Voltage output	--	--	1.4	--	V
	Loading current	DCVDD=1.4V@IOVDD = 3.3V, LDO mode	--	--	60	mA
		DCVDD=1.4V@VBAT = 3.7V, DCDC mode	--	--	180	mA
Low Power mode						

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
IOVDD	Loading current	IOVDD=3.0V@VBAT = 3.7V	--	--	10	mA

3.4 IO Characteristics

Table 3-4 IO Characteristics

Input Characteristics						
Symbol	Parameter	Conditions	IO	Min	Max	Unit
V_{IL}	Low-Level Input Voltage	IOVDD = 3.0V	PA5~PA7 PB2~PB7 PC2~PC10 PC13, PD8 PD11~PD15 PE0~PE5 PF0~PF5 HUSBDP HUSBDM PR0~PR1	-0.3	1.4	V
V_{IH}	High-Level Input Voltage	IOVDD = 3.0V	PA5~PA7 PB2~PB7 PC2~PC10 PC13, PD8 PD11~PD15 PE0~PE5 PF0~PF5 HUSBDP HUSBDM PR0~PR1	1.7	3.3	V
Output Characteristics						
Symbol	Parameter	Conditions	IO	Typ	Unit	
$ I_{OL} $	Output Current	IOVDD = 3.0V Voutput = 0.3V	PA5~PA7 PB2~PB7 PC2~PC10 PC13, PD8 PD11~PD15 PE0~PE5 PF0~PF5	2.5(HD=0) 8(HD=1) 18.5(HD=2) 24(HD=3)	mA	
			PR0~PR1	2.5(HD=0) 18.5(HD=1)	mA	
			HUSBDP HUSBDM	8	mA	
$ I_{OH} $	Output Current	IOVDD = 3.0V Voutput = 2.7V	PA5~PA7 PB2~PB7	2.5(HD=0) 8(HD=1)	mA	

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
			PC2~PC10 PC13, PD8 PD11~PD15 PE0~PE5 PF0~PF5	18.5(HD=2) 24(HD=3)		
			PR0~PR1	2.5(HD=0) 18.5(HD=1)		mA
			HUSBDP HUSBDM	8		mA
Internal Resistance Characteristics						
Symbol	Parameter	Conditions	IO	Typ		Unit
R _{pu}	Pull-up Resistance	IOVDD = 3.0V	PA5~PA7 PB2~PB7 PC2~PC10 PC13, PD8 PD11~PD15 PE0~PE5 PF0~PF5	10k		Ω
			HUSBDP	1.5k(PU=1) 1k(PU=2/3)		Ω
R _{pd}	Pull-down Resistance	IOVDD = 3.0V	PA5~PA7 PB2~PB7 PC2~PC10 PC13, PD8 PD11~PD15 PE0~PE5 PF0~PF5	10k		Ω
			HUSBDP HUSBDM	15k		Ω

Note

1.Internal pull-up/pull-down resistance accuracy ±20%.

3.5 Audio DAC Characteristics

Table 3-5 Stereo DAC Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Resolution	--	--	16	--	bits
Output Sample Rate	--	8	--	96	kHz
SNR	Differential Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	103	--	dB
	Single Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	100	--	dB
Dynamic Range	Differential Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	103	--	dB
	Single Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	100	--	dB
THD+N	Differential Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	-77	--	dB
	Single Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted load=10kΩ	--	-64.8	--	dB
Noise Floor	Differential Mode B/W=20Hz~20kHz A-Weighted load=10kΩ	--	6.4	--	uVrms
	Single Mode B/W=20Hz~20kHz A-Weighted load=10kΩ	--	5.4	--	uVrms

3.6 Audio ADC Characteristics

Table 3-6 Audio ADC Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Resolution	--	--	16	--	bits
Input Sample Rate	--	8	--	48	kHz
SNR	Differential input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	95	--	dB
	Single-ended input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	92	--	dB
Dynamic Range	Differential input Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	95	--	dB
	Single-ended input Mode Fin=1kHz@-60dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	92	--	dB
THD+N	Differential input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	-87	--	dB
	Single-ended input Mode Fin=1kHz@0dBFS Fs=44.1kHz B/W=20Hz~20kHz A-Weighted ADC gain=0dB	--	-81	--	dB
Analogue Gain		-6	--	28	dB
Max Input Level	Differential input Mode ADC gain=0dB	--	0.7	--	Vrms
	Single-ended input Mode ADC gain=0dB	--	0.35	--	Vrms

3.7 BT Characteristics

3.7.1 Transmitter

Table 3-7-1 Transmitter characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Maximum RF Transmit Power	BR	--	19	--	dBm
Maximum RF Transmit Power	EDR $\pi/4$ DQPSK EDR 8DPSK	--	19	--	dBm
Relative Transmit Power	EDR $\pi/4$ DQPSK EDR 8DPSK	--	1.5	--	dB
Maximum RF Transmit Power	BLE-1Mbps	--	19	--	dBm

3.7.2 Receiver

Table 3-7-2 Receiver characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Sensitivity	BR	--	-95.5	--	dBm
	EDR $\pi/4$ DQPSK	--	-96	--	dBm
	EDR 8DPSK	--	-88	--	dBm
	BLE-1Mbps	--	-98	--	dBm
	BLE-2Mbps	--	-95	--	dBm
	BLE-S2	--	-101	--	dBm
	BLE-S8	--	-106	--	dBm

3.8 WiFi Characteristics

3.8.1 Transmitter

TX Power with Spectral Mask and EVM Meeting 802.11 Standards.

Table 3-8-1 Transmitter characteristics

Parameter	Conditions	Min	Typ	Max	Unit
TX Power	802.11b, 1 Mbps, DSSS	--	19	--	dBm
	802.11b, 11 Mbps, CCK	--	19	--	dBm
	802.11g, 6 Mbps, OFDM	--	17	--	dBm
	802.11g, 54 Mbps, OFDM	--	14	--	dBm
	802.11n, HT20, MCS0	--	17	--	dBm
	802.11n, HT20, MCS7	--	13	--	dBm
	802.11n, HT40, MCS0	--	17	--	dBm
	802.11n, HT40, MCS7	--	12	--	dBm

3.8.2 Receiver

For RX tests, the PER (packet error rate) limit is 8% for 802.11b, and 10% for 802.11g/n.

Table 3-8-2 Receiver characteristics

Parameter	Conditions	Min	Typ	Max	Unit
-----------	------------	-----	-----	-----	------

Parameter	Conditions	Min	Typ	Max	Unit
Sensitivity	802.11b, 1 Mbps, DSSS	--	-97	--	dBm
	802.11b, 2 Mbps, DSSS	--	-94.5	--	dBm
	802.11b, 5.5 Mbps, CCK	--	-93	--	dBm
	802.11b, 11 Mbps, CCK	--	-90	--	dBm
	802.11g, 6 Mbps, OFDM	--	-93	--	dBm
	802.11g, 9 Mbps, OFDM	--	-92	--	dBm
	802.11g, 12 Mbps, OFDM	--	-91	--	dBm
	802.11g, 18 Mbps, OFDM	--	-89	--	dBm
	802.11g, 24 Mbps, OFDM	--	-86	--	dBm
	802.11g, 36 Mbps, OFDM	--	-83	--	dBm
	802.11g, 48 Mbps, OFDM	--	-79	--	dBm
	802.11g, 54 Mbps, OFDM	--	-77	--	dBm
	802.11n, HT20, MCS0	--	-93	--	dBm
	802.11n, HT20, MCS1	--	-90.5	--	dBm
	802.11n, HT20, MCS2	--	-88	--	dBm
	802.11n, HT20, MCS3	--	-84.5	--	dBm
	802.11n, HT20, MCS4	--	-81.5	--	dBm
	802.11n, HT20, MCS5	--	-77	--	dBm
	802.11n, HT20, MCS6	--	-75	--	dBm
	802.11n, HT20, MCS7	--	-74	--	dBm
	802.11n, HT40, MCS0	--	-89	--	dBm
	802.11n, HT40, MCS1	--	-86	--	dBm
	802.11n, HT40, MCS2	--	-84	--	dBm
	802.11n, HT40, MCS3	--	-80	--	dBm
	802.11n, HT40, MCS4	--	-77.5	--	dBm
	802.11n, HT40, MCS5	--	-72.5	--	dBm
	802.11n, HT40, MCS6	--	-71.5	--	dBm
	802.11n, HT40, MCS7	--	-70	--	dBm
	802.11n, HT40, MCS32	--	-89	--	dBm

4 Package Information

4.1 QFN68_7*7mm

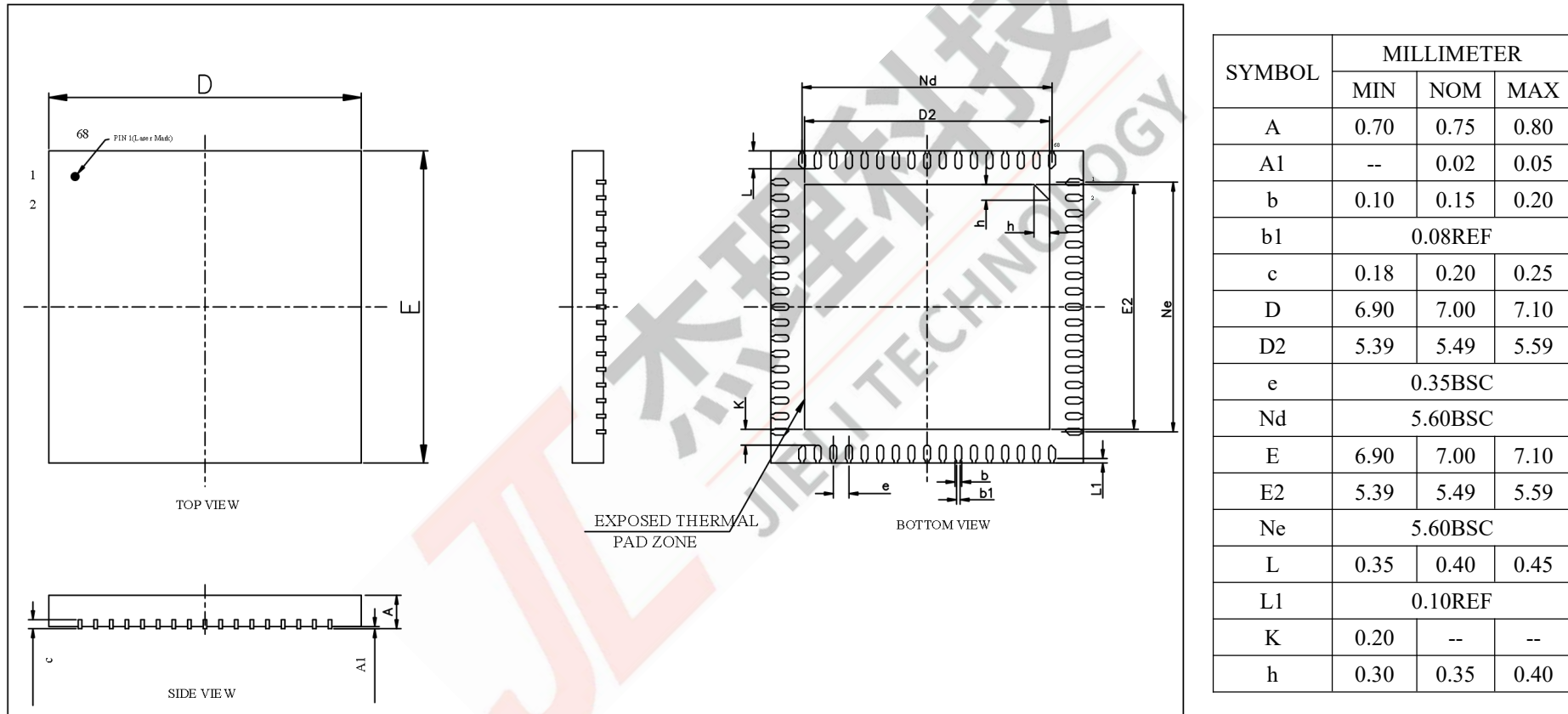


Figure 4-1 AC7925A Package

5 IC Marking Information

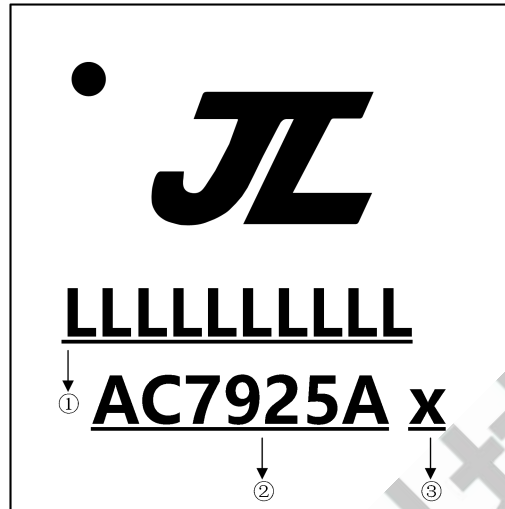


Figure 5-1 AC7925A Package Outline

- ① Production Batch
- ② Chip Model
- ③ Built-in DDR size
 - 0 No Flash Memory
 - 2 2Mbit flash
 - 4 4Mbit flash
 - 8 8Mbit flash
 - 6 16Mbit flash
 - 3 32Mbit flash
 - 5 64Mbit flash
 - 7 128Mbit flash
 - A 1Mx16 SDRAM
 - B 4Mx16 SDRAM
 - E 4Mx16bit DDR1
 - F 8Mx16bit DDR1
 - G 16Mx16bit DDR1

6 Solder-Reflow Condition

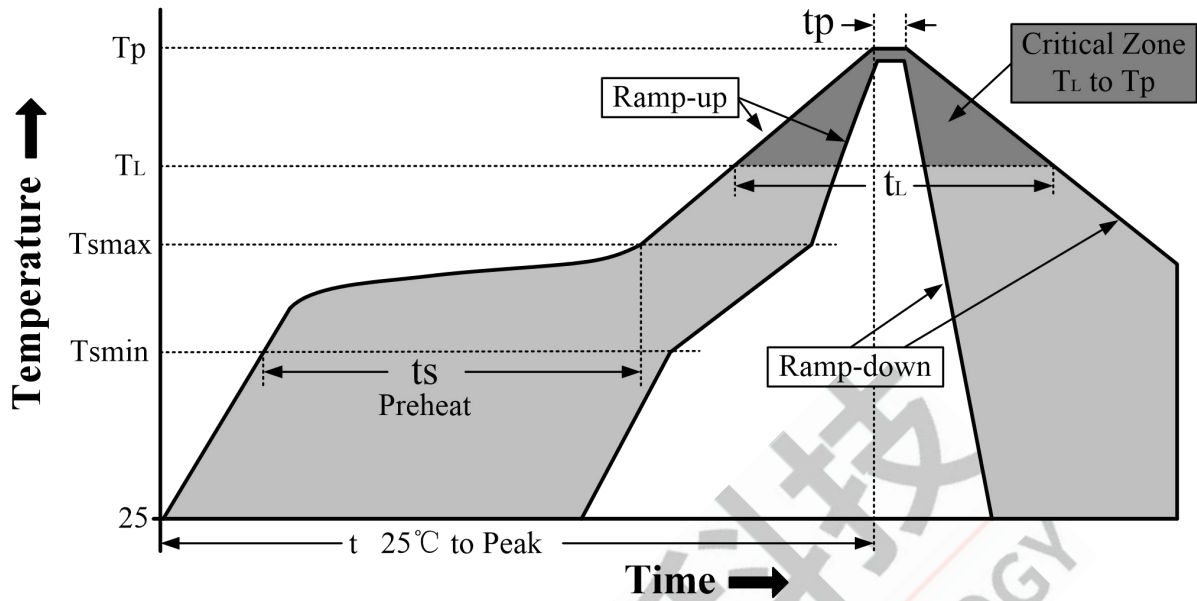


Figure 6-1 Classification Reflow Profile

Table 6-1 Classification Profiles

Profile Feature		Sn-Pb Eutectic Assembly	Pb-Free Assembly
Preheat/Soak	Temperature Min (T_{smin})	100°C	150°C
	Temperature Max (T_{smax})	150°C	200°C
	Time (t_s) from (T_{smin} to T_{smax})	60-120 seconds	60-180 seconds
Average ramp-up rate (T_{smax} to T_p)		3°C/second max	3°C/second max
Liquidous temperature (T_L)		183°C	217°C
Time (t_L) maintained above T_L		60-150 seconds	60-150 seconds
Peak package body temperature (T_p)		See Table 6-2	See Table 6-3
Time within 5°C of actual Peak Temperature (t_p) ²		10-30 seconds	20-40 seconds
Ramp-down rate (T_p to T_L)		6°C/second max	6°C/second max
Time 25°C to peak temperature		6 minutes max	8 minutes max

Note

- 1.All temperatures refer to topside of the package, measured on the package body surface
- 2.Time within 5°C of actual peak temperature (t_p) specified for the reflow profiles is a "supplier" and "user" maximum.

Table 6-2 SnPb Classification Temperature

Package Thickness	Volume mm ³ < 350	Volume mm ³ ≥ 350
<2.5 mm	240 +0/-5°C	225 +0/-5°C
≥2.5 mm	225 +0/-5°C	225 +0/-5°C

Table 6-3 Pb-free - Classification Temperature

Package Thickness	Volume mm ³ < 350	Volume mm ³ 350 - 2000	Volume mm ³ > 2000
< 1.6mm	260°C	260°C	260°C
1.6 mm - 2.5mm	260°C	250°C	245°C
> 2.5mm	250°C	245°C	245°C

Note

1.*Tolerance The device manufacturer/supplier shall assure process compatibility up to and including the stated classification temperature (this means Peak reflow temperature +0°C.For example 260°C+0°C)at the rated MSL level.